

Ramon Costa

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EDUCATION

University of Washington

Bachelor of Science in Computer Engineering; Minor in Mathematics

Seattle, WA

Sept. 2023 – Jun. 2026

EXPERIENCE

Systems Team Consultant

Jan. 2024 – Present

UW Academic Technologies

Seattle, WA

- Administer and support 1,000+ macOS and 500+ Windows systems across campus, ensuring optimal uptime and system reliability.
- Developed and deployed a Linux lab environment utilizing Bash scripting and Active Directory, establishing a new university-wide standard adopted into the official IT handbook.
- Contributed to sprint planning and infrastructure migrations within an Agile framework, enhancing system reliability and ensuring seamless service continuity for 1,500+ campus devices.
- Diagnose and resolve complex technical issues, utilizing asset and ticket data to optimize system performance across distributed workstations.

Teaching Assistant — Fundamentals of Computer Science II

Mar. 2023 – Jun. 2023

Bellevue College

Bellevue, WA

- Facilitated weekly virtual labs and office hours for 36 students, breaking down complex concepts like recursion, inheritance, and polymorphism into accessible lessons.
- Guided the design and implementation of efficient data structures, emphasizing time and space complexity analysis ($O(n)$) for searching and sorting algorithms.
- Provided constructive code reviews on student projects, helping improve their algorithmic thinking, code readability, and runtime performance.

PROJECTS

AstraPath — Full-Stack Route Optimizer | *Next.js, TypeScript, Neo4j, Clerk, Vercel*

- Engineered a graph-based supply chain optimizer using Neo4j AuraDB to model complex distribution networks and calculate Pareto-dominant paths across cost, speed, and CO_2 emissions.
- Implemented dynamic Cypher queries with weighted edge logic to simulate real-time market conditions like fuel price fluctuations and weather congestion.
- Architected a multi-tenant isolation layer ensuring secure, private workspaces for authenticated users via Clerk OAuth and session-based guest access.
- Developed an interactive data visualization canvas using react-force-graph, enabling real-time node manipulation and persistent layout restoration.

DUET — Social Music Networking App | *Dart, Flutter, Firebase (Auth, Firestore, Messaging), Spotify API*

- Led a team using Agile methodologies, facilitating weekly sprint planning and stand-up sessions to coordinate feature development.
- Developed a web app that analyzes Spotify API data to match users based on musical compatibility and listening habits.
- Engineered a real-time messaging system using Firestore, ensuring low-latency communication and synchronized data states across clients.
- Architected a scalable backend using Firebase Auth and Firestore, implementing secure user data handling and efficient query structures.

TECHNICAL SKILLS

Languages: Python, Java, C#, C++, JavaScript, TypeScript, SQL, HTML, CSS, Dart

Frameworks & Libraries: React, Next.js, Node.js, Express.js, Flutter, PyTorch, JUnit, Unity, Tailwind CSS

Databases: Neo4j (Graph), MySQL, SQLite, Firebase Firestore

Cloud & DevOps: Microsoft Azure, Vercel, Docker, GitHub Actions, CI/CD, Linux, Bash, PowerShell

Developer Tools: Git (GitHub, GitLab), Clerk Auth, Jamf, Maven, GDB, VS Code, Jira

Additional Skills: RESTful APIs, Agile Methodologies, Web Security (XSS/CSRF), Data Structures & Algorithms, Figma